Lec-62

2 → omp

3 → MST

99 %

Numerical

Lec_61

3. Huffman Coding

Greedy
1. Knapsack prob
2. Job sequence problem

- encoding → converting data into secret format

a = 11, b = 01, c = 00, d = 10

01101

b d a

Huffman Coding is an algorithm used for lossless data compression.

- It is Used for Data Compression ,Data Encoding
- Non-uniform Technic => each data has different bits.
- More frequency ->less bits (less number of bits
- Less frequency -> more bits (more no of bits)

a = 50, b = 20, c = 3

less bits

a = 01, c = 01100

ab => 16 bit

1001 4 bit

ab

a = 110, b = 01

abc 5

abc

a → 8 bit

10 2 bit

b → 8 bit

01

a=45, b=1,c=3, d=20,e=30,f=1 using Huffman coding find

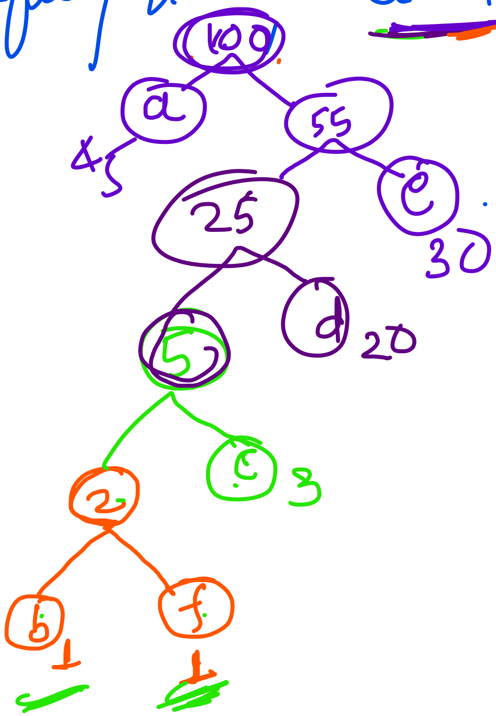
1)Code assigned for each Character

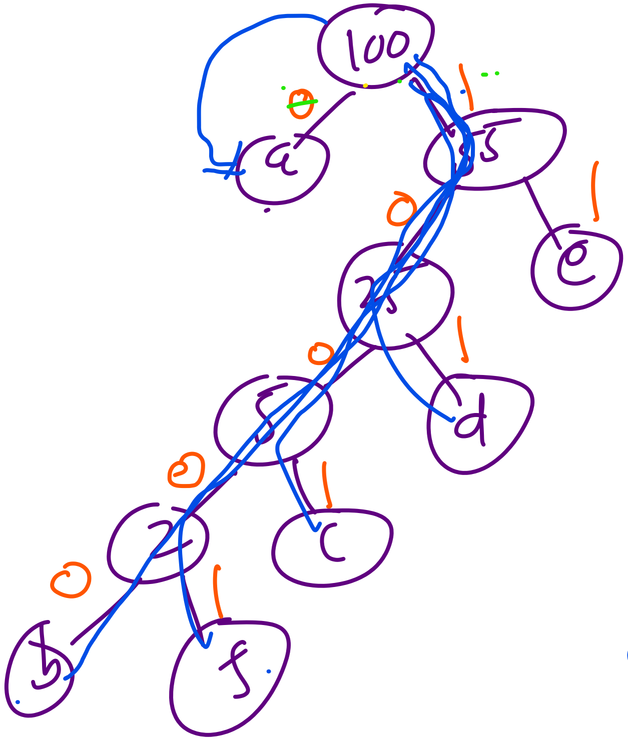
2) No bits per Character

solve:- frequency given

a=45, ~~b=1~~, ~~c=3~~, ~~d=20~~, ~~e=30~~, ~~f=1~~, ~~2~~, ~~5~~, ~~25~~

55





Left = 0
Right = 1

① Start from root

a = 0 , b = 10000

c = 1001

d = 101

e = 11

f ⇒ 10001

2 no. of bits for each char

a = 45, b = 1, c = 3, d = 20, e = 30, f = 1

no of bits: a = 1, b = 5, c = 4, d = 3, e = 2, f = 5

Total char = 100

no. of Total bits = $45\times1 + 1\times5 + 3\times4 + 20\times3 + 30\times2 + 1\times5$

$= 45 + 5 + 12 + 60 + 60 + 5$

$= 50 + 12 + 120 + 5$

$= 67 + 120$

$= 187$

2) 1.87 bits/char

no of bits per char =

for 100 char => 187 bits.

1 char = $\frac{187}{100}$ = 1.87

given data: a = 45, ~~b = 1~~, c = 20, d = 3, e = 30, f = 1 O(1)

① take two min from given data

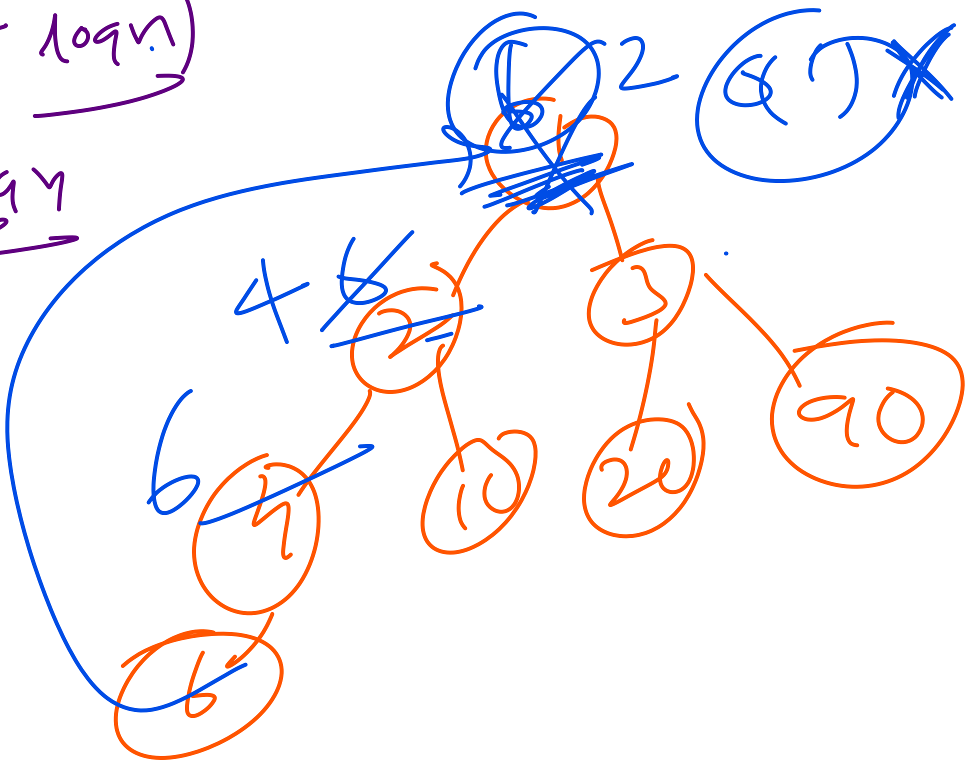
n inputs

Algorithm:

- Create a min heap for the given frequency using build heap method ⇒ O(n)
- Take to minimum from the heap and delete them 2 log n
  two
- Create a new Huffman encoded tree make above two minimums leaf of O(1) that tree and add both minimums make parent of two minimums the result log n
- Insert result into min heap log n
- Repeat this procedure until no node is present in min heap (n-1) times
- TC=? ⇒ O(n) + (n-1)(2 log n + log n)

TC = n + (n-1) 2 log n + n log n

TC = n log n



$T_c = n\log n$ each Imp

min Heap

→ Data str. use :-